

## **HOMEWORK: K-Nearest Neighbors (KNN) on Diabetes Dataset**

Course: Computational Science for Computer Science

### **Topic: Machine Learning – K-Nearest Neighbors**

#### **Objective:**

This homework aims to develop your understanding of instance-based learning by applying the K-Nearest Neighbors (KNN) algorithm to a real-world dataset. You will analyze, preprocess, and classify data to predict diabetes outcomes.

#### **Dataset Description:**

You are provided with a dataset containing **768 patient records** with the following features:

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function
- Age
- Outcome (Target Variable: 0 = Non-diabetic, 1 = Diabetic)

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**Part 1: Data Understanding**

| Feature                           | Description                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------|
| <b>Pregnancies</b>                | Number of times pregnant. High values can correlate with gestational diabetes risk. |
| <b>Glucose</b>                    | Plasma glucose concentration. Usually the most significant predictor of diabetes.   |
| <b>Blood Pressure</b>             | Diastolic blood pressure (mm Hg). High BP is often comorbid with diabetes.          |
| <b>Skin Thickness</b>             | Triceps skin fold thickness (mm), used to estimate body fat.                        |
| <b>Insulin</b>                    | 2-Hour serum insulin (mu U/ml). Zero values here are often missing data.            |
| <b>BMI</b>                        | Body Mass Index (weight in kg/(height in m)). An indicator of obesity.              |
| <b>Diabetes Pedigree Function</b> | A score that indicates the genetic probability of diabetes based on family history. |
| <b>Age</b>                        | Age in years. Risk typically increases with age.                                    |

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| Feature | Description                                      |
|---------|--------------------------------------------------|
| Outcome | Class variable (0 = Non-diabetic, 1 = Diabetic). |

### 1. Which features are likely most important for prediction?

In predicting diabetes, the most important features are those that have a high physiological correlation with blood sugar regulation and metabolic health.

- **Glucose:** This is the most critical feature. Since diabetes is defined by the body's inability to manage blood sugar, a high plasma glucose concentration is the primary indicator. In a KNN algorithm, a test instance with high glucose will naturally be "closer" in distance to other diabetic patients.
- **BMI (Body Mass Index):** Obesity is a major risk factor for Type 2 diabetes. High BMI often correlates with insulin resistance. In the dataset, BMI provides a strong "clustering" signal—patients with higher BMIs are more likely to be grouped near diabetic outcomes.
- **Age:** The risk of developing Type 2 diabetes generally increases as people age due to metabolic changes. It serves as an important demographic weight in the KNN distance calculation.
- **Diabetes Pedigree Function:** This feature is important because it represents the genetic component. It scores the likelihood of diabetes based on family history; a higher score significantly shifts the patient's "location" in the multi-dimensional space toward a diabetic classification.

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### 2. Which features may contain missing or problematic values?

Problematic values in this dataset are usually "0" entries in columns where a zero is biologically impossible for a living person. These are not true zeros; they are placeholders for missing data.

- **Insulin:** This is the most problematic column. In this dataset, nearly half the records have a "0" for Insulin. A human cannot have a zero insulin level while alive. These "0" values act as "noise"—if you don't fix them, the KNN algorithm will calculate distances based on these fake zeros, leading to incorrect neighbor matching.

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- **Skin Thickness:** Similar to Insulin, a significant number of rows show "0" for skin thickness. This is physically impossible and indicates that the measurement was not taken or recorded for that patient.
- **Blood Pressure & BMI:** Some records show "0" for Blood Pressure or BMI. These are clearly errors in data collection, as a patient must have a weight and a pulse to be part of the study.
- **Glucose:** While rare (only 5 instances), even Glucose has "0" values. These must be addressed because Glucose is your most important predictor; having a "0" there will drastically throw off the accuracy of your KNN model.

## Part 2: Data Preprocessing

**Table 1: Missing and Zero Value Identification**

This table explains why we need to preprocess the data. It shows the number of "0" values that were treated as missing and then successfully filled.

| Feature Name  | Count of Zeros (Raw Data) | Count of Zeros (After Imputation) | Imputation Method |
|---------------|---------------------------|-----------------------------------|-------------------|
| Glucose       | 5                         | 0                                 | Median Imputation |
| BloodPressure | 35                        | 0                                 | Median Imputation |
| SkinThickness | 227                       | 0                                 | Median Imputation |
| Insulin       | 374                       | 0                                 | Median Imputation |

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| Feature Name | Count of Zeros (Raw Data) | Count of Zeros (After Imputation) | Imputation Method |
|--------------|---------------------------|-----------------------------------|-------------------|
| <b>BMI</b>   | 11                        | 0                                 | Median Imputation |

**Table 2: Summary Statistics (Before vs. After Preprocessing)**

This table demonstrates that we successfully applied **Standardization**. In the "After" columns, a Mean of ~0 and a Std Dev of 1 prove the data is now correctly scaled for the KNN algorithm.

| Feature               | Mean (Original) | Std Dev (Original) | Mean (Scaled) | Std Dev (Scaled) |
|-----------------------|-----------------|--------------------|---------------|------------------|
| <b>Pregnancies</b>    | 3.85            | 3.37               | 0.00          | 1.00             |
| <b>Glucose</b>        | 120.89          | 31.97              | 0.00          | 1.00             |
| <b>Blood Pressure</b> | 69.11           | 19.36              | 0.00          | 1.00             |
| <b>Skin Thickness</b> | 20.54           | 15.95              | 0.00          | 1.00             |
| <b>Insulin</b>        | 79.80           | 115.24             | 0.00          | 1.00             |
| <b>BMI</b>            | 31.99           | 7.88               | 0.00          | 1.00             |

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| Feature  | Mean<br>(Original) | Std Dev<br>(Original) | Mean<br>(Scaled) | Std Dev<br>(Scaled) |
|----------|--------------------|-----------------------|------------------|---------------------|
| Pedigree | 0.47               | 0.33                  | 0.00             | 1.00                |
| Age      | 33.24              | 11.76                 | 0.00             | 1.00                |

### Part 3: KNN Implementation

#### 1. The Mathematical Foundation: Euclidean Distance

The K-Nearest Neighbors (KNN) algorithm classifies a data point by calculating how far away it is from every other point in the training set. We use **Euclidean Distance** (d), which is the straight-line distance between two points in a multi-dimensional space.

The formula is:

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

#### 2. Step-by-Step Manual Solving Process

##### Step A: Define the Test Instance

We selected **Test Instance Index 668** from the dataset. After preprocessing (Standardization), its coordinates are:

- **Pregnancies:** 0.640 | **Glucose:** -0.778 | **Blood Pressure:** -1.190 | **Skin Thickness:** 0.443
- **Insulin:** 0.571 | **BMI:** 0.225 | **Pedigree:** -0.126 | **Age:** 0.830

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**Step B: Detailed Calculation (Comparing Test 668 to Train 535)**

We find the difference for every feature, square it, sum them, and take the square root.

| Feature (i)         | Test Value (xi) | Train Value (yi) | Difference (x-y) | Squared Difference (x-y) <sup>2</sup> |
|---------------------|-----------------|------------------|------------------|---------------------------------------|
| 1. Pregnancies      | 0.640           | 0.046            | 0.594            | 0.353                                 |
| 2. Glucose          | -0.778          | 0.340            | -1.118           | 1.250                                 |
| 3. BloodPressure    | -1.190          | -0.032           | -1.158           | 1.341                                 |
| 4. SkinThickness    | 0.443           | -0.012           | 0.455            | 0.207                                 |
| 5. Insulin          | 0.571           | -0.182           | 0.753            | 0.567                                 |
| 6. BMI              | 0.225           | 0.065            | 0.160            | 0.026                                 |
| 7. Pedigree         | -0.126          | -0.513           | 0.387            | 0.150                                 |
| 8. Age              | 0.830           | -0.871           | 1.701            | 2.893                                 |
| <b>Sum (Σ)</b>      |                 |                  |                  | <b>6.787</b>                          |
| <b>Distance (d)</b> |                 |                  |                  | $\sqrt{6.787} = 2.605$                |

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3. Sorting and Finding Neighbors

After performing the calculation above for 10 training samples, we sort them by distance (smallest to largest). The smallest distances represent the "Nearest Neighbors."

| Rank | Train Index | Euclidean Distance | Actual Outcome (Class) |
|------|-------------|--------------------|------------------------|
| 1st  | 535         | 2.605              | 1 (Diabetic)           |
| 2nd  | 718         | 2.686              | 0 (Non-diabetic)       |
| 3rd  | 60          | 2.729              | 0 (Non-diabetic)       |
| 4th  | 306         | 3.143              | 1                      |
| 5th  | 90          | 3.445              | 0                      |
| 6th  | 231         | 3.581              | 1                      |
| 7th  | 346         | 3.651              | 0                      |
| 8th  | 618         | 3.775              | 1                      |
| 9th  | 340         | 3.879              | 0                      |



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| Rank | Train Index | Euclidean Distance | Actual Outcome (Class) |
|------|-------------|--------------------|------------------------|
| 10th | 294         | 3.945              | 0                      |

To classify Test Instance 668, we apply the **Majority Voting** rule using the K=5 nearest neighbors:

- **Neighbor 1:** Outcome 1
- **Neighbor 2:** Outcome 0
- **Neighbor 3:** Outcome 0
- **Neighbor 4:** Outcome 1
- **Neighbor 5:** Outcome 0
- **Result:** Three "0"s vs. two "1". The final predicted class is **0 (Non-diabetic)**.

### Overall Model Evaluation

To validate the model's effectiveness, the KNN algorithm was tested across the entire dataset using different values for K.

| K-Value | Accuracy | Observation                                                                           |
|---------|----------|---------------------------------------------------------------------------------------|
| K=3     | 70.78%   | High sensitivity to local noise in the dataset.                                       |
| K=5     | 73.38%   | <b>Optimal Performance:</b> Provides the best balance of accuracy and generalization. |

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| K-Value | Accuracy | Observation                                                             |
|---------|----------|-------------------------------------------------------------------------|
| K=7     | 71.43%   | Performance begins to drop as the decision boundary becomes too smooth. |

#### Part 4: Model Evaluation

Based on the experimental runs with the diabetes dataset using an 80/20 train-test split, the model achieved its highest performance at **K=5**.

| K Value    | Accuracy Percentage  |
|------------|----------------------|
| K=3        | 70.78%               |
| <b>K=5</b> | <b>73.38% (Best)</b> |
| K=7        | 71.43%               |

**Confusion Matrix (for K=5):** A Confusion Matrix shows how many patients were correctly or incorrectly classified.

- **True Negatives (TN):** 81 (Correctly predicted non-diabetic)
- **True Positives (TP):** 32 (Correctly predicted diabetic)
- **False Positives (FP):** 18 (Healthy patients incorrectly flagged as diabetic)
- **False Negatives (FN):** 23 (Diabetic patients incorrectly flagged as healthy)

**Question 1: Which value of K performed best?**

In our testing, **K=5** performed best with an accuracy of **73.38%**. This value provided the best balance between capturing relevant patterns and ignoring random noise in the data.

**Question 2: Why does performance change with different K values?**

Performance changes because K determines the size of the "neighborhood" used for classification.

- KNN is an instance-based learner that relies on local density.
- When K changes, the "boundary" between diabetic and non-diabetic regions shifts.
- If the neighborhood is too small, the model reacts to every single outlier.
- If it is too large, the model includes points that are too far away and irrelevant to the specific test instance.

**Question 3: What happens when K is too small or too large?**

- **When K is too small (Overfitting):** The model becomes overly sensitive to "noise" or errors in the training data. For example, if a healthy patient was mislabeled as diabetic, a small K would let that single error incorrectly influence any test points nearby.
- **When K is too large (Underfitting):** The model becomes too "blunt." It starts to ignore local similarities and simply predicts based on the majority class of the entire dataset. This causes the model to miss subtle patterns that distinguish diabetic from non-diabetic patients in specific metabolic clusters.

## Part 5: Analysis and Reflection

The K-Nearest Neighbors (KNN) algorithm is a non-parametric, instance-based learning method that classifies data points based on their proximity to others in a multi-dimensional space. One of the primary **strengths of KNN** is its simplicity and intuitive nature. Unlike other models that require complex training phases, KNN is a "lazy learner," meaning it stores the training data and performs computations only at the time of prediction. This makes it highly flexible, as it can naturally handle multi-class problems and adapt quickly as new data is added to the system without retraining the entire model. Furthermore, since it makes no underlying assumptions about the distribution of the data, it is effective in scenarios where the decision boundary is irregular or non-linear.

However, the algorithm has significant **limitations**, particularly regarding **computational cost**. As the dataset grows, the time required to calculate the distance between a test instance and every single training sample increases linearly, making it inefficient for very large datasets. Another critical weakness is its extreme **sensitivity to scaling**. Because KNN relies on Euclidean distance, features with larger numerical ranges (like Insulin or Glucose) would dominate the distance calculation if not standardized, leading to biased predictions. The algorithm is also sensitive to irrelevant features or "noise," such as the biologically impossible "0" values identified in our dataset, which can significantly skew the neighborhood selection if left unaddressed.

**KNN is appropriate** for low-dimensional datasets where the local structure of the data is a strong indicator of the class. It is particularly useful in medical diagnosis support where similar patient profiles (neighbors) often share similar health outcomes. Conversely, it is **not appropriate** for high-dimensional data (the "curse of dimensionality") or datasets where feature scales vary wildly and cannot be normalized.

Reflecting on the **observations from this experiment**, the importance of data quality was paramount. We observed that features like **Glucose and BMI** acted as primary clustering signals. The manual computation for Test Instance 668 demonstrated that even a single outlier or a lack of standardization would have shifted the "nearest" neighbor from a correct match to an incorrect one. The experiment highlighted that  $K=5$  was the "sweet spot" for this dataset; it was large enough to filter out local noise but small enough to maintain the distinct characteristics of the local patient clusters. This balance resulted in a final accuracy of **73.38%**, proving that even a

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simple distance-based model can provide meaningful insights when paired with rigorous preprocessing.

### Graphs & Visualization:

